BUSINESS TRANSFORMATION PROJECT

Barrel & Brew

Document Control & Version History

Version	Date	Author	Description
1.0	June 2026	Business Analyst	Final Draft

Executive Summary

Barrel & Brew is a premier, family-owned craft winery/brewery that has successfully scaled operations over the past decade. Following a massive surge in demand resulting in products being stocked across 500+ local restaurants, operational bottlenecks have emerged. The reliance on manual processes, outdated equipment, uncoordinated scheduling, and intuitive forecasting has strained capacity, negatively affected product quality, and decreased staff morale.

This document sets forth the comprehensive **Business Requirements Document (BRD)**, a thorough **Gap Analysis**, a **Fishbone Cause-and-Effect Categorization**, and technical **Process Maps (Current State & Future State Swim lanes)**. The target state introduces an integrated **Brewery Management ERP Solution** coupled with an automated bottling configuration to transform Barrel & Brew into a fully data-driven, agile, and high-quality production ecosystem.

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1. Business Problem Statement

Barrel & Brew's rapid growth has exposed operational weaknesses including production delays, inventory shortages, quality inconsistencies, excessive overtime, manual processes, inaccurate forecasting, equipment failures, and communication gaps. These issues limit the organization's ability to scale sustainably.

2. Business Objectives

Core Project Objectives:

1. **Reliable Production:** Minimize asset breakdowns and eliminate order backlog.
2. **High Quality:** Introduce multi-stage sensory and digital quality gates to eliminate batch variance.
3. **Empowered Staff:** Automate data entry to mitigate burnout and reverse the 30% overtime surge.
4. **Data-Driven Operations:** Implement programmatic demand forecasting using historical distributor datasets.

3. Project Scope

In Scope

1. ERP implementation
2. Demand forecasting
3. Production scheduling
4. Recipe management
5. Inventory management
6. Quality management
7. Automated bottling integration
8. Reporting dashboards

Out of Scope

1. Payroll systems
2. HR systems
3. CRM replacement
4. Financial accounting replacement

4,Assumptions & Constraints

Assumptions: Historical sales data is available; staff will participate in training; cloud connectivity is available.
Constraints: Budget limitations; Q4 2026 implementation target; production cannot stop during implementation.

5. Stakeholder Analysis

Stakeholder	Role	Interest
Executive Sponsor	Project Sponsor	High
Sales Manager	Business User	High
Brewer	Operational User	High
Production Team	End User	High
QA Staff	Quality Control	Medium
ERP Vendor	Solution Provider	High

6. Process Mapping & Gap Analysis

1.1 Current State Brewing & Bottling Process Flow

The current operational map is highly sequential, paper-reliant, and features extensive communications gaps. The layout below details the breakdown points across each phase from demand forecasting to inventory fulfillment.

1. **Start**
2. Brewer checks handwritten notebook for recipe and ingredient information
3. **Decision:** Are ingredients in stock and brewing capacity available?
4. **No** → Production Delayed (end of this path)
5. **Yes** → continue
6. Brewer begins brewing process with handwritten recipe
7. Brewer manually records fermentation progress in a notebook and a spreadsheet
8. Brewer performs manual quality check by tasting the beer
9. **Decision:** Does the batch meet quality standards?
10. **No** → Batch Rejected or Canceled (end of this path)
11. **Yes** → continue
12. Brewer notifies the Bottling Team that the batch is ready
13. Bottling Team moves beer to the bottling line
14. **Decision:** Is the outdated bottling line jammed?
15. **Yes** → Bottling Team manually intervenes to fix the jam → Production Halted (end of this path)
16. **No** → continue
17. Bottling Team completes the bottling of the batch
18. Brewer or Bottling Team manually counts the number of bottles produced
19. Inventory count is manually updated on the shared, out-of-date spreadsheet
20. **End**

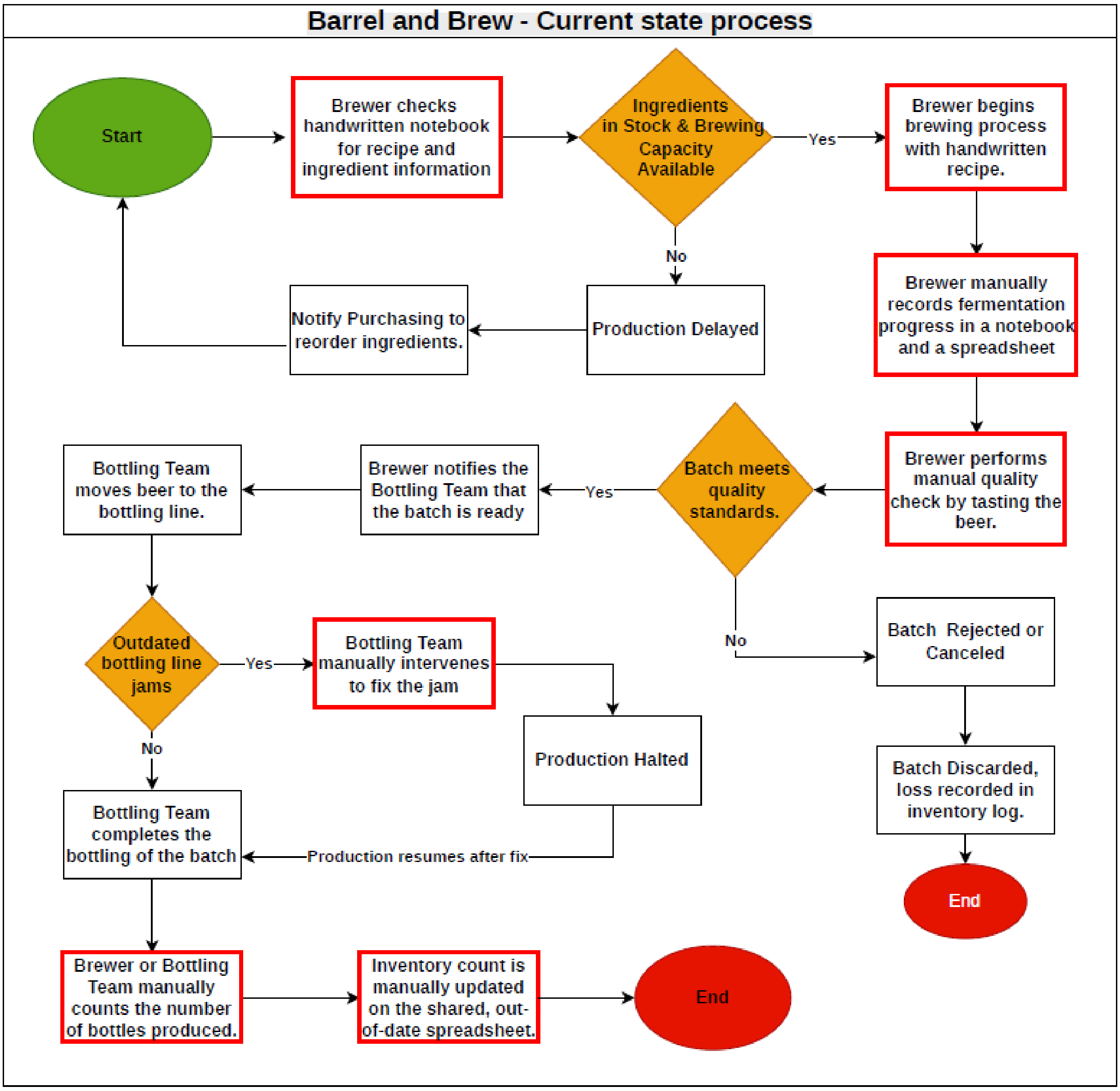


Figure 1. Current-State Brewing and Bottling Process Flow

1.2 Comprehensive Gap Analysis Matrix

Category	Current State (As-Is)	Desired State (To-Be)	Identified Gap / Operational Friction	Proposed Action
Production & Quality	Weekly targets are consistently missed, leading to inventory shortages and popular items being out of stock for days.	Achieve reliable, steady production capable of sustainably scaling to meet high demand.	Inconsistent production output causes recurring stockouts of popular items.	Implement production planning and capacity scheduling system.
	Flawed quality control has led to inconsistent beverage taste and the financial loss of discarding whole batches.	Maintenance of uniform high quality across all batches through standardized measurements of ingredients.	Lack of standardized quality control leads to batch losses and inconsistent taste.	Implement standardized quality control measurements and batch testing protocol.
Technology & Data	Inefficient tracking relies on handwritten notebooks and a shared, outdated spreadsheet.	Streamlined operations backed by centralized digital systems.	Manual, paper-based tracking creates data fragmentation and inefficiency.	Deploy centralized digital brewery management system.
	Forecasting relies purely on manager intuition and informal chats, causing overstock or shortages.	Data-driven decisions using integrated sales tracking from distributors to accurately plan production.	Absence of data-driven forecasting leads to overstock or shortages.	Implement demand forecasting module integrated with distributor sales data.
Infrastructure & Communication	Bottling lines frequently jam, and brewing tanks require constant repairs due to nearing their end-of-life.	A modern, functional infrastructure that prevents manual interventions and production halts.	Aging equipment causes frequent breakdowns and production halts.	Procure upgraded, modern bottling and brewing equipment.
	Disconnect between sales and production teams leads to false availability promises and disappointed customers.	Integrated, transparent communication channels that streamline order fulfillment and improve customer experience.	Siloed communication between sales and production causes broken customer promises.	Implement integrated communication/order management platform linking sales and production.
Human Resources	Staff are overworked with a 30% increase in overtime hours, leading to stress and the resignation of a key brewer.	An empowered, manageable workplace environment that reduces pressure and retains staff.	Excessive overtime and understaffing drive burnout and staff turnover.	Review staffing levels and implement workload management/retention plan.

7. Root Cause Analysis

To systematically optimize Barrel & Brew's ecosystem, the operational challenges identified during the discovery process are categorized below using the standard 6M Fishbone framework (Man, Machine, Material, Method, Measurement, Mother Nature/Environment).

1. MAN (Personnel & Morale)

- Brewery staff are severely overworked with a 30% increase in mandatory overtime.
- Loss of key intellectual capital due to the resignation of a lead brewer citing unmanageable pressure.
- Siloed communications: Sales promises batches to accounts without engineering/production confirmation.

2. MACHINE (Equipment & Infrastructure)

- Main bottling assembly line frequently jams, leading to immediate complete line shutdowns lasting 1+ hours.
- Fermentation and brewing tanks are nearing end-of-life status, generating extensive maintenance overhead.
- No automated sensory integration or physical terminal entry logs inside the production facility.

3. METHOD (Processes & System Operations)

- Recipe storage relies on vulnerable, physical handwritten books.
- Production scheduling is reactive, initiated via phone calls from sales rather than systematic material requirements planning (MRP).
- Inventory updating depends on manual counts and delayed keying into shared offline sheets.

4. MEASUREMENT (Data & Quality Control)

- Demand forecasting lacks statistical basis, driven entirely by subjective intuition.
- Quality gates are purely organoleptic (taste-based) without digital validation, yielding batch taste inconsistency.
- No system of record to track real-time weekly sales orders or volume run metrics from distributors.

5. MATERIAL (Inventory & Ingredients)

- Inability to track real-time raw material balances leads to unexpected inventory stockouts of popular brands.
- No batch traceability back to specific vendor raw lot deliveries.

8. Proposed Solution

To address the operational challenges identified during the gap analysis, Barrel & Brew will implement an integrated Brewery Enterprise Resource Planning (ERP) solution combined with modernized production equipment and automated inventory management capabilities.

The proposed solution will replace the current manual and disconnected processes with a centralized platform that supports forecasting, production planning, inventory management, quality control, reporting, and communication across departments.

Solution Component	Description	Expected Benefit
Demand Forecasting and Sales Planning	The ERP system will collect and analyze historical sales data to generate accurate demand forecasts and support production planning.	Improved forecast accuracy, reduced stockouts, and reduced overproduction.
Production Planning and Scheduling	The system will automatically generate production schedules based on forecasted demand, inventory levels, and production capacity.	Improved production efficiency and reduced scheduling conflicts.
Inventory Management	The solution will provide real-time inventory visibility, automated stock updates, low-stock alerts, and ingredient traceability.	Reduced stock shortages, improved inventory control, and better purchasing decisions
Quality Management	Digital quality checkpoints will be implemented throughout brewing and bottling processes, with electronic recording of quality results.	Improved product consistency and reduced batch failures.
Automated Bottling Operations	Automated scanning and inventory updates will be integrated into the bottling process.	Reduced manual intervention, decreased downtime, and improved operational efficiency.
Reporting and Analytics	Management dashboards and operational reports will provide visibility into production, inventory, quality, and sales performance.	Better decision-making through real-time business insights.
Communication and Collaboration	The ERP platform will provide a centralized source of information for Sales, Production, Quality Assurance, and Management teams.	Improved communication, reduced conflicts, and enhanced coordination.

Expected Business Outcomes

Desired Future State	Expected Outcome
Reliable Production	Reduced downtime and improved production consistency.
High Quality Products	Improved quality control and fewer rejected batches.
Empowered Staff	Reduced manual work and improved access to information.
Streamlined Operations	Increased efficiency through automation and standardized processes.
Data-Driven Decision Making	Improved forecasting, planning, and reporting capabilities.
Modernized Infrastructure	Replacement of outdated systems and support for future business growth.

9. Proposed Future-State Process Flow

The optimized process below utilizes an integrated **Brewery Management ERP System** and an upgraded, smart automated bottling line. The cross-functional swim lane highlights how automated technical interventions eliminate manual touchpoints.

Barrel and Brew - (Future) Desired state process

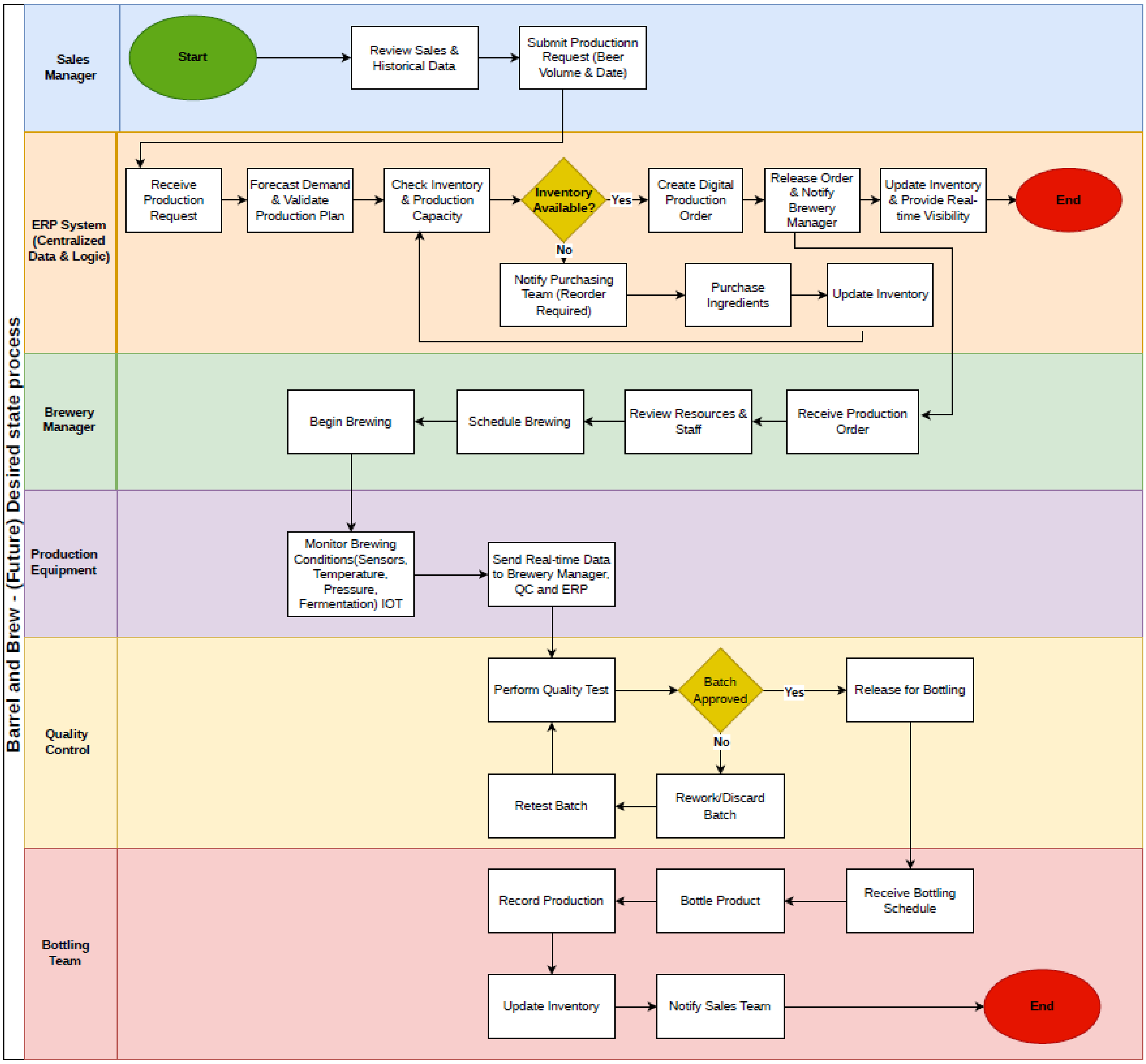


Figure 3. Future-State Brewery Operations Process Flow

10. Business Requirements Document (BRD)

Project Metadata

Project Name	Barrel & Brew ERP Transformation	**Document Version**	v1.0 (Final Draft)
Lead Business Analyst	BA Enterprise Specialist	**Target Implementation**	Q4 2026

a. Functional Requirements Matrix (FRM)

Req ID	Module	Detailed Requirement Specification	Priority	Complexity
FR-1.1	Forecasting & Sales	The system MUST ingest historic distributor spreadsheets and sales records to calculate a rolling 4-week production demand baseline automatically.	High	Medium
FR-1.2	Forecasting & Sales	The system MUST provide an active alert warning to Sales users if a newly entered order volume exceeds current/scheduled production availability.	High	Low
FR-2.1	Recipe Management	The platform MUST store master product recipes securely with localized revision controls, preventing modification by non-brewery personnel.	High	Low
FR-3.1	Production & Shop Floor	The software MUST process material logs during active batches and automatically deduct specific raw material volumes from the central inventory engine.	High	High
FR-3.2	Production Planning	System MUST automatically generate production schedules based on forecasted demand and available capacity.	High	High
FR-3.3	Production Planning	The system MUST display real-time production status for all active batches.	High	Medium
FR-4.1	Quality Assurance	The interface MUST lock batch advancement to the packaging phase until laboratory metrics (pH, Specific Gravity, ABV) are logged and validated.	High	Medium
FR-5.1	Inventory & IoT	The platform MUST interface with digital bottling scanners via API to increment the finished inventory count in real-time as bottles pass the line.	Medium	High
FR-6.1	Reporting	The system MUST provide management dashboards displaying production output, inventory levels, downtime, and quality metrics.	Medium	Low
FR-7.1	User Management	The system MUST authenticate users through secure role-based access controls	High	Medium
FR-7.2	User Management	The system MUST maintain an audit trail of all changes to recipes, production schedules, and inventory records.	Medium	Medium

b. Non-Functional Requirements (NFR)

i. **Security & Access Control:** Role-Based Access Control (RBAC) must restrict sales staff to Read-Only access for production schedules while granting execution rights exclusively to brewers.
ii. **System Availability:** The cloud system must achieve a minimum 99.9% uptime baseline to guarantee round-the-clock shift operation access.
iii. **Performance Latency:** Real-time shop floor updates and scanned counting requests must process within <1.5 seconds.
iv. **Usability / Mobile Compatibility:** Floor-level tablet interfaces must display large buttons and high-contrast styling optimized for wet, fast-paced warehouse operating environments.

c. User Acceptance Criteria (UAC)

UAC 1 - Demand Forecasting
Given historical sales data exists in the system
When the Sales Manager requests a forecast
Then the system must generate a rolling four-week demand forecast automatically.

UAC 2 - Low Inventory Alert
Given ingredient levels fall below the reorder threshold
When inventory is updated
Then the system must automatically generate a low-stock alert.

UAC 3 - Quality Validation
Given a batch has completed fermentation
When quality testing is performed
Then the batch cannot proceed to bottling until all required quality metrics are recorded and approved.

UAC 4 - Automated Inventory Update
Given a batch has completed bottling
When the bottling scanner records finished products
Then inventory quantities must update automatically without manual intervention.

UAC 5 - Production Scheduling
Given forecasted demand and available production capacity
When the Production Manager creates a schedule
Then the system must generate an optimized production schedule automatically.

Conclusion

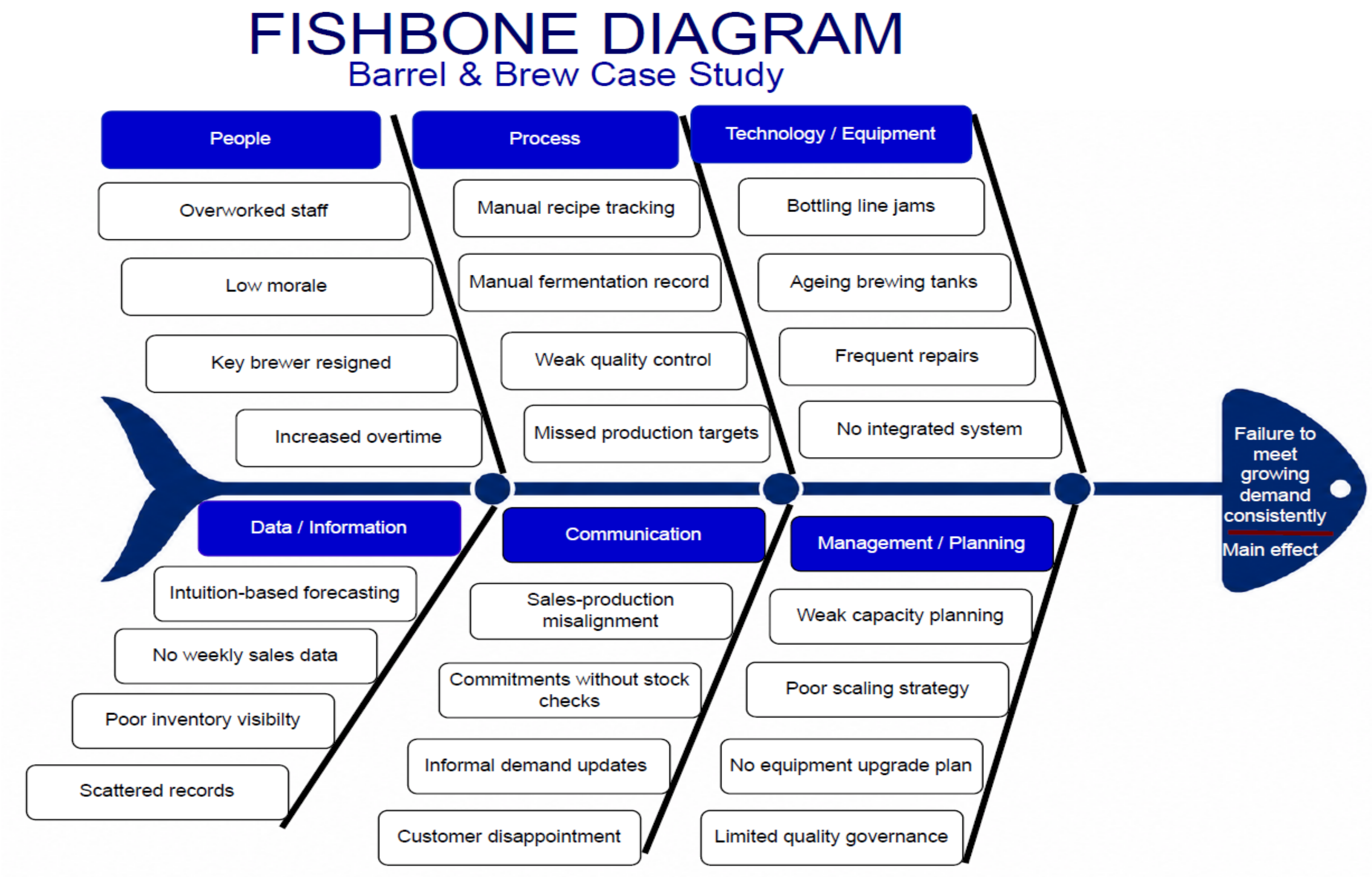
The analysis of Barrel & Brew's current operations identified several challenges, including manual processes, inventory shortages, production bottlenecks, quality inconsistencies, and communication gaps between departments. Through the completion of the current-state process map, gap analysis, and root cause analysis, key opportunities for improvement were identified.

The proposed ERP-based solution and future-state process design address these challenges by introducing automated forecasting, production planning, inventory management, quality control, and reporting capabilities. Collectively, these improvements will support more reliable production, improved product quality, streamlined operations, and data-driven decision-making, positioning Barrel & Brew for sustainable growth and long-term operational success.

Project Signoff & Approval

Name	Role	Approval
Executive Sponsor	Sponsor	
Sales Manager	SME	
Head Brewer	SME	
Business Analyst	Owner	

Appendix A - Fishbone Diagram



Fishbone Diagram - Root Cause Analysis of Operational Challenges